Q1. Explain working of POP protocol.

The Post Office Protocol version 3 (POP3) is an application layer protocol used for retrieving emails from a mail server to a client device. It is mainly designed for offline email access, where emails are downloaded and stored on the user's local system.

The working of POP3 is divided into three important phases:

• Authorization Phase
In this phase, the client establishes a connection with the mail server using TCP on port 110 (or port 995 for secure communication). The user then sends login credentials such as username and password. The server verifies these details and grants access to the mailbox if the credentials are correct.

• Transaction Phase
After successful authentication, the client enters the transaction phase. In this stage, the user can check the number of emails, retrieve specific messages, and mark emails for deletion. It is important to note that messages marked for deletion are not removed immediately.

• Update Phase
This phase starts when the client terminates the session by sending the quit command. At this point, all messages that were marked for deletion are permanently removed from the server.

POP3 follows a download-and-delete approach, where emails are transferred from the server to the client device and typically removed from the server. This makes it efficient but unsuitable for accessing emails from multiple devices.

Q2. Compare OSI model with TCP/IP model.

Feature	OSI Model	TCP/IP Model
Full Form	Open Systems Interconnection	Transmission Control Protocol/Internet Protocol
Layers	7	4
Developed by	ISO	DARPA
Development	Protocol developed after model	Model developed after protocols
Approach	Vertical	Horizontal
Reliability	Less reliable	More reliable
Usage	Reference/Theoretical model	Practical/Implementation model
Transport Layer	Guarantees delivery	Does not guarantee delivery
Session & Presentation	Separate layers	Merged into Application layer

Q3. Explain protocols working at each layer in TCP/IP model.

The TCP/IP model is a four-layer model used for data communication over the Internet. Each layer performs specific functions, and different protocols operate at each layer to ensure proper communication between devices.

The layers and their associated protocols are explained below:

• **Application Layer**

This layer provides services directly to the user and enables interaction with network applications. It combines the functions of application, presentation, and session layers of the OSI model. Protocols at this layer include HTTP for web communication, FTP for file transfer, SMTP for sending emails, POP3 and IMAP for receiving emails, and DNS for domain name resolution.

• **Transport Layer**
This layer is responsible for end-to-end communication, reliability, flow control, and error handling. The main protocols used are TCP and UDP. TCP provides reliable, connection-oriented communication with error checking and retransmission, while UDP provides fast, connectionless communication without reliability.

• **Internet Layer**
This layer handles logical addressing and routing of data packets across networks. The main protocol is IP, which is responsible for delivering packets from source to destination. Other protocols include ICMP, which is used for error reporting and diagnostics, and ARP, which is used to map IP addresses to physical MAC addresses.

• **Network Interface Layer**
This is the lowest layer of the TCP/IP model. It is responsible for transmitting data over the physical network. It includes protocols and technologies such as Ethernet and Wi-Fi, which handle framing, physical addressing, and actual data transmission over the medium.

Q4. Briefly explain OSI model with all its layers and functionality of each layer.

The OSI (Open Systems Interconnection) model is a reference model developed by the International Organization for Standardization to standardize communication between different systems. It consists of seven layers, where each layer performs a specific function in the process of data communication.

The layers of the OSI model and their functionalities are explained below:

• **Physical Layer**
This is the lowest layer of the OSI model. It is responsible for the transmission of raw bits over the physical medium. It deals with hardware elements such as cables, switches, voltage levels, and data transmission rates.

• **Data Link Layer**
This layer ensures reliable transmission of data over a physical link. It performs framing, error detection, and flow control. It also handles physical addressing using MAC addresses.

• **Network Layer**
The network layer is responsible for logical addressing and routing of data packets from source to destination across multiple networks. It determines the best path for data transmission.

• **Transport Layer**
This layer provides end-to-end communication between devices. It ensures reliable data

transfer, error correction, and flow control. It also divides data into segments and reassembles them at the destination.

• **Session Layer**
The session layer is responsible for establishing, maintaining, and terminating communication sessions between applications. It manages synchronization and checkpoints during data transfer.

• **Presentation Layer**
This layer is responsible for data representation and formatting. It handles encryption, decryption, compression, and conversion of data into a readable format.

• **Application Layer**
This is the topmost layer and provides services directly to the user. It allows applications to access network services such as file transfer, email, and web browsing.

Q5. Give the difference between ARP and RARP protocols.

Basis	**ARP (Address Resolution Protocol)**	**RARP (Reverse Address Resolution Protocol)**
Full Form	Address Resolution Protocol	Reverse Address Resolution Protocol
Purpose	Finds MAC address using IP address	Finds IP address using MAC address
Working Direction	IP → MAC	MAC → IP
Usage	Used in normal network communication	Used by diskless systems at startup
Current Status	Widely used in modern networks	Obsolete and rarely used
Server Requirement	Does not require dedicated server (uses broadcast)	Requires a RARP server
Replacement	Still in use	Replaced by DHCP

Q6. Explain working of IMAP protocol.

The Internet Message Access Protocol (IMAP) is an application layer protocol used to access and manage emails directly on the mail server. Unlike POP3, IMAP allows users to view and organize emails without downloading them permanently, making it suitable for accessing email from multiple devices.

The working of IMAP is explained through the following phases:

- **Connection Establishment**

The email client establishes a connection with the mail server using TCP, typically on port 143 or port 993 for secure communication using SSL/TLS.

- **Authentication Phase**

The user logs in by providing a username and password. The server verifies the credentials and grants access to the mailbox.

- **Mailbox Selection**

After successful login, the client selects a mailbox (such as Inbox) to access emails stored on the server.

- **Email Access and Management**

The user can read, search, organize, and manage emails directly on the server. Emails are not downloaded permanently unless explicitly requested. Actions like deleting or moving emails are reflected on the server.

- **Synchronization**

IMAP maintains synchronization between the server and multiple client devices. Any changes made on one device are automatically updated on others.

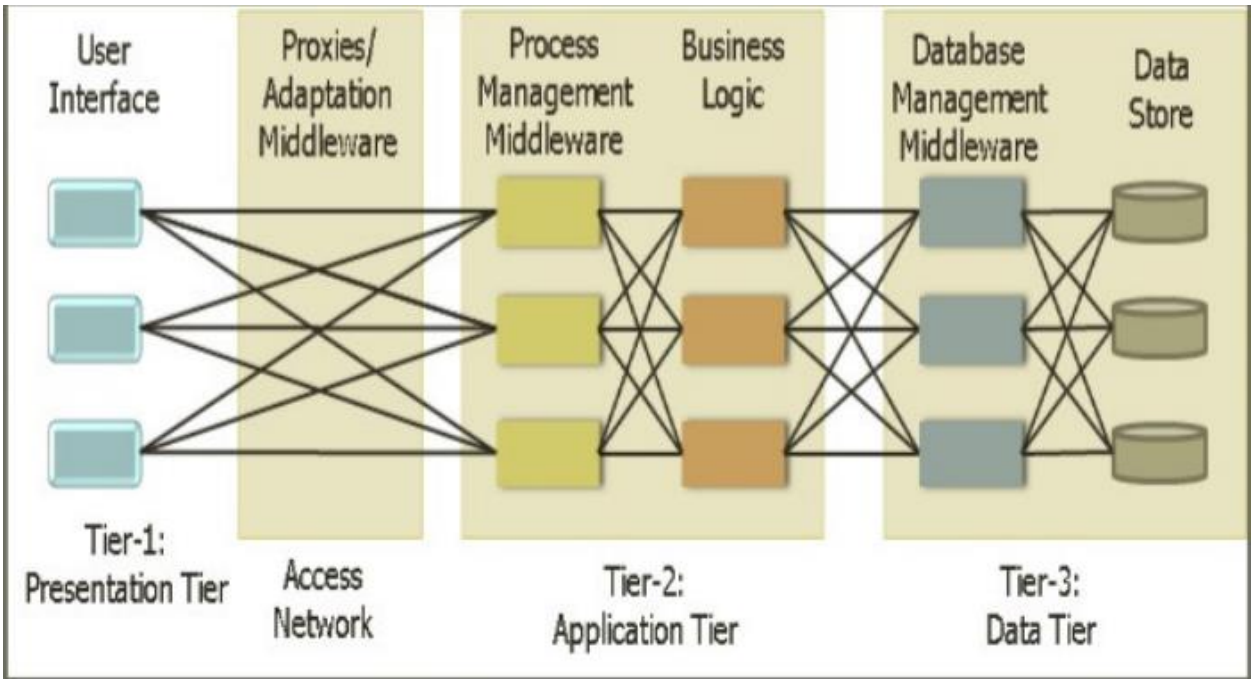
- **Termination Phase**

When the user logs out, the session ends, but all changes remain stored on the server.

IMAP works on a server-based model, where emails are stored and managed centrally. This makes it more flexible and suitable for modern usage compared to POP3.

Q7. Explain Three-tier architecture of mobile computing with appropriate diagram.

The three-tier architecture of mobile computing is a design model that divides a mobile system into three independent layers. This separation improves scalability, flexibility, and maintainability of mobile applications.



The three tiers are explained below:

• **Presentation Tier (Client Layer)**
This is the front-end layer where the user interacts with the system. It includes mobile devices such as smartphones, tablets, or laptops. This layer is responsible for displaying the user interface and sending user requests to the server. It does not perform heavy processing.

• **Application Tier (Middle Layer / Server Layer)**
This layer acts as a bridge between the client and the database. It processes user requests, applies business logic, and controls data flow. It is usually implemented using application servers. This layer ensures that the system functions correctly and efficiently.

• **Data Tier (Database Layer)**
This is the back-end layer where data is stored and managed. It includes database servers that handle data storage, retrieval, and updates. This layer ensures data consistency and security.

Q9. Explain Advantages of IPv6 over the older IPv4 addressing scheme.

The advantages of IPv6 over IPv4 are explained below:

• **Larger Address Space**
IPv6 uses 128-bit addressing, which provides a very large number of IP addresses compared to IPv4's 32-bit addressing. This solves the problem of address exhaustion.

• **Better Security**
IPv6 has built-in support for security through IPsec, which provides authentication and encryption of data packets.

• **Simplified Header Format**
The IPv6 header is simpler and more efficient than IPv4, which improves packet processing speed and routing efficiency.

• **Improved Routing Efficiency**
IPv6 reduces the size of routing tables and supports hierarchical addressing, making routing faster and more efficient.

• **Auto Configuration**
IPv6 supports automatic address configuration (stateless autoconfiguration), allowing devices to configure themselves without needing a DHCP server.

• **No Need for NAT (Network Address Translation)**
Due to the large address space, IPv6 eliminates the need for NAT, allowing direct communication between devices.

• **Better Support for Mobility and Multicasting**
IPv6 provides improved support for mobile devices and efficient data transmission using multicast instead of broadcast.

Q10. Enlist types of networks available in mobile computing. Explain one of them in detail.

The types of networks available in mobile computing are as follows:

• Wireless Personal Area Network (WPAN)
• Wireless Local Area Network (WLAN)
• Wireless Metropolitan Area Network (WMAN)
• Wireless Wide Area Network (WWAN)

Wireless Local Area Network (WLAN)

A Wireless Local Area Network (WLAN) is a network that allows devices to connect and communicate wirelessly within a limited geographical area such as a home, office, or campus. It is commonly based on Wi-Fi technology.

WLAN operates using radio waves instead of physical cables. Devices such as laptops, smartphones, and tablets connect to a central device called an access point. This access point connects the wireless devices to a wired network or the Internet.

The working of WLAN involves a wireless device sending signals to the access point. The access point then forwards the data to the desired destination, either within the same network or to an external network like the Internet.

WLAN provides flexibility and mobility, allowing users to move freely within the coverage area while staying connected. It is easy to install and reduces the need for extensive cabling. However, it may face issues such as signal interference and security risks if not properly configured.

Q11. Explain types of Routing.

Routing is the process of selecting the best path for data to travel from a source to a destination across a network. It is performed by network devices called routers.

The main types of routing are explained below:

- **Static Routing**

In static routing, the path for data transmission is manually defined by the network administrator. The router does not change the path automatically. It is simple to implement but not suitable for large or dynamic networks because it cannot adapt to network changes.

- **Dynamic Routing**

In dynamic routing, routers automatically determine the best path using routing protocols. These protocols update routing tables based on network conditions such as traffic and link failures. It is more flexible and efficient for large networks.

- **Default Routing**

Default routing is used when the destination network is not known. In this case, the router forwards the data to a predefined default path. It is commonly used in small networks or for connecting to the Internet.

- **Adaptive Routing**

Adaptive routing is a type of dynamic routing where routing decisions are adjusted automatically based on current network conditions like congestion and failures. It improves performance by selecting optimal paths dynamically.

- **Non-Adaptive Routing**

In non-adaptive routing, the routing path is fixed and does not change according to network conditions. It is simple but inefficient when network conditions vary.

Q12. What is Subnetting and Supernetting?

Subnetting and supernetting are techniques used in IP addressing to manage and organize networks efficiently.

- **Subnetting**

Subnetting is the process of dividing a large network into smaller sub-networks called subnets. It is done by borrowing bits from the host portion of an IP address to create multiple smaller networks. Subnetting improves network performance, enhances security, and allows better utilization of IP addresses. It also reduces network congestion by limiting broadcast domains.

- **Supernetting**

Supernetting is the process of combining multiple smaller networks into a single larger network. It is also known as route aggregation or CIDR (Classless Inter-Domain Routing). In this method, bits are borrowed from the network portion to create a larger address block. Supernetting helps in reducing the size of routing tables and improves routing efficiency.

Q13. Explain IPv6 Addressing. Why need of IPv6 migration?

IPv6 addressing is the scheme used to identify devices in a network using the Internet Protocol version 6. It was developed to overcome the limitations of IPv4, especially the shortage of IP addresses.

IPv6 uses 128-bit addresses, which are written in hexadecimal format and separated by colons. A typical IPv6 address consists of eight groups of four hexadecimal digits. This large address space allows a virtually unlimited number of devices to be connected to the Internet.

An IPv6 address is generally divided into two main parts:

- **Network Prefix**

This part identifies the network to which the device belongs.

- **Interface Identifier**

This part identifies a specific device within that network.

IPv6 also supports different types of addresses such as unicast (one-to-one communication), multicast (one-to-many communication), and anycast (one-to-nearest communication).

Need for IPv6 Migration

The transition from IPv4 to IPv6 is necessary due to several reasons:

• **Exhaustion of IPv4 Addresses**
IPv4 uses 32-bit addressing, which provides a limited number of addresses. With the rapid growth of Internet users and devices, these addresses are no longer sufficient.

• **Better Security**
IPv6 includes built-in support for security features like IPsec, which ensures secure communication.

• **Improved Performance**
IPv6 has a simplified header structure, which improves routing efficiency and packet processing speed.

• **Elimination of NAT**
IPv6 removes the need for Network Address Translation due to its large address space, enabling direct device-to-device communication.

• **Auto Configuration**
IPv6 supports automatic configuration of devices without requiring manual setup or DHCP.

• **Support for Modern Applications**
IPv6 is better suited for mobile computing, IoT devices, and real-time applications.

Q15. Write short note on Network Address Translation (NAT).

Network Address Translation (NAT) is a technique used in networking to modify IP address information in packet headers while data is transmitted across a router or firewall. It is mainly used to allow multiple devices in a private network to share a single public IP address.

NAT works by translating the private IP addresses of devices inside a local network into a public IP address when accessing external networks such as the Internet. When a device sends a request, the NAT-enabled router replaces the private IP address with its own public IP address and keeps a record of this mapping. When the response is received, the router uses this record to forward the data back to the correct device inside the network.

The important types of NAT are as follows:

• **Static NAT**
In static NAT, a single private IP address is mapped to a single public IP address. This mapping remains constant and is mainly used when a device needs to be accessed from outside the network.

• **Dynamic NAT**
In dynamic NAT, private IP addresses are mapped to a pool of public IP addresses. The assignment is done dynamically based on availability.

• **Port Address Translation (PAT)**
Also known as NAT overload, this method allows multiple devices to share a single public IP address by using different port numbers. It is the most commonly used form of NAT.

Q17. Explain working of Indirect TCP.

Indirect TCP is a technique used in mobile computing to improve the performance of TCP connections when a mobile host is involved. It separates the connection into two parts to handle issues caused by mobility, such as disconnections and varying network conditions.

The working of Indirect TCP can be understood as follows:

• **Connection Splitting**
In Indirect TCP, a single TCP connection is divided into two separate connections. One connection exists between the fixed host and a base station (or access point), and the other connection exists between the base station and the mobile host.

• **Fixed Host to Base Station Communication**
The connection between the fixed host and the base station uses standard TCP. This part of the connection is stable because it operates over a wired network.

• **Base Station to Mobile Host Communication**
The second connection, between the base station and the mobile host, is managed separately. This link is wireless and may experience frequent disconnections or errors.

• **Data Transfer Process**
When the fixed host sends data, it is first received by the base station. The base station acknowledges the data to the fixed host and then forwards it to the mobile host. This improves performance because the fixed host does not need to wait for the mobile host's acknowledgment.

• **Handling Disconnections**
If the mobile host becomes temporarily disconnected, the base station can store the data and retransmit it once the connection is restored. This prevents unnecessary retransmissions from the fixed host.

• **Limitation**
One major drawback is that it breaks end-to-end TCP semantics because acknowledgments are sent by the base station instead of the mobile host.

Q19. Explain Communication Middleware in detail.

Communication middleware is a software layer that acts as an intermediary between different applications or systems in a distributed or mobile computing environment. It enables communication and data exchange between heterogeneous systems without requiring them to understand each other's internal workings.

The role of communication middleware is to hide the complexity of the underlying network and provide a uniform interface for communication.

The main functions of communication middleware are as follows:

• **Abstraction of Communication**
It hides low-level networking details such as protocols, data formats, and transmission mechanisms, allowing developers to focus on application logic.

• **Interoperability**
It enables communication between different systems, platforms, and programming languages, ensuring that heterogeneous systems can work together.

• **Message Handling**
Middleware manages the sending, receiving, and formatting of messages between systems. It ensures that data is delivered correctly and efficiently.

• **Location Transparency**
It allows applications to communicate without knowing the physical location of other systems in the network.

• **Scalability Support**
Middleware supports the addition of new devices and services without affecting existing systems.

• **Security Services**
It provides features such as authentication, authorization, and encryption to ensure secure communication.

• **Reliability and Fault Tolerance**
Middleware ensures that communication continues even in case of failures by using mechanisms like message buffering and retransmission.

Common types of communication middleware include Remote Procedure Call (RPC), Message-Oriented Middleware (MOM), and Object Request Brokers (ORB).

Q20. Explain Handover management in Mobile IP.

Handover management in Mobile IP refers to the process of maintaining continuous communication when a mobile node moves from one network to another. Since the IP address of a device is linked to its network location, movement across networks can disrupt communication. Handover management ensures that the connection remains active without interruption.

The working of handover management involves the following key steps:

• **Movement Detection**
When a mobile node moves to a new network, it detects the change using agent advertisements or solicitations from foreign agents. This helps the node identify that it is no longer in its home network.

• **Care-of Address Acquisition**
After detecting movement, the mobile node obtains a new temporary address called the care-of address. This address represents the current location of the mobile node in the foreign network.

• **Registration Process**
The mobile node registers its care-of address with its home agent. This allows the home agent to know the current location of the mobile node and update its records accordingly.

• **Packet Forwarding**
Once registration is complete, the home agent intercepts packets sent to the mobile node's home address and forwards them to the care-of address using tunneling.

• **Handover Execution**
During movement between networks, the connection is maintained by quickly updating the care-of address and forwarding mechanism. This ensures minimal disruption in communication.

• **Types of Handover**
Handover can be classified as hard handover, where the old connection is broken before a new one is established, and soft handover, where both connections exist temporarily to ensure smooth transition.

Handover management is essential in mobile computing to provide seamless connectivity and uninterrupted service as users move across different networks.

Q21. Explain key functions of Communication Gateways.

A communication gateway is a network device or software component that connects different networks and enables communication between systems that may use different protocols, formats, or architectures. It plays a crucial role in mobile computing by ensuring smooth data exchange across heterogeneous environments.

The key functions of communication gateways are as follows:

- **Protocol Conversion**
The gateway converts data from one protocol to another so that systems using different communication standards can interact with each other.

- **Data Format Translation**
It translates data formats between different systems, ensuring that information is correctly interpreted at both ends.

- **Message Routing**
The gateway determines the correct path for data and forwards messages to the appropriate destination network or device.

- **Security Enforcement**
It provides security features such as authentication, encryption, and firewall protection to ensure safe communication between networks.

- **Data Filtering**
The gateway can filter unwanted or unauthorized data, allowing only valid and necessary information to pass through.

- **Connection Management**
It establishes, maintains, and terminates communication sessions between different systems.

- **Interoperability Support**
The gateway enables communication between heterogeneous systems, such as mobile devices, servers, and legacy systems, making them work together seamlessly.

Q22. Explain Process of Mobile IP.

Mobile IP is a protocol that allows a mobile device to move from one network to another while maintaining its permanent IP address. This ensures continuous communication without interruption.

The process of Mobile IP works through the following steps:

- **Agent Discovery**
When a mobile node enters a network, it detects whether it is in its home network or a foreign network using advertisements from home agents or foreign agents.

- **Care-of Address Assignment**
If the mobile node is in a foreign network, it obtains a temporary address called the care-of address. This address represents the current location of the mobile node.

- **Registration**
The mobile node registers its care-of address with its home agent. This process may involve the foreign agent, which forwards the registration request to the home agent. The home agent stores this information.

- **Packet Interception**
When a packet is sent to the mobile node, it is first received by the home agent because the sender uses the mobile node's permanent home address.

- **Tunneling**
The home agent encapsulates the original packet inside a new packet and forwards it to the care-of address. This process is called tunneling.

- **Packet Delivery**
The foreign agent (or the mobile node itself) receives the tunneled packet, decapsulates it, and delivers it to the mobile node.

- **Communication Continuity**
Even when the mobile node moves again, the process repeats, ensuring that communication remains uninterrupted.

Q23. List advantages of WPANs.

Wireless Personal Area Networks (WPANs) are short-range wireless networks used for communication between personal devices such as smartphones, laptops, and wearable devices.

The advantages of WPANs are as follows:

• **Low Power Consumption**
WPAN devices consume very little power, making them suitable for battery-operated devices.

• **Cost Effective**
They are inexpensive to set up and maintain since they do not require complex infrastructure.

• **Easy Installation**
WPANs are simple to install and configure, with minimal technical requirements.

• **Mobility Support**
They allow users to move freely within a short range while staying connected.

• **Cable-Free Communication**
WPANs eliminate the need for physical cables, reducing clutter and improving convenience.

• **Flexible Connectivity**
Multiple devices such as phones, laptops, and headsets can easily connect and communicate.

• **Short Range Security**
Due to limited coverage area, WPANs provide better security as access is restricted to nearby devices.

Q24. Explain steps of packet delivery in Mobile IP.

Packet delivery in Mobile IP ensures that data sent to a mobile node reaches it correctly even when it is away from its home network. This is achieved using home agents, foreign agents, and care-of addresses.

The steps involved in packet delivery are as follows:

• **Packet Sent by Correspondent Node**
A sender (correspondent node) sends a packet to the mobile node using its permanent home IP address. The sender is not aware of the mobile node's current location.

• **Packet Received by Home Agent**
The packet reaches the home network, where the home agent intercepts it because the mobile node is not currently present there.

• **Encapsulation (Tunneling)**
The home agent encapsulates the original packet inside a new IP packet. This new packet contains the care-of address of the mobile node as the destination.

• **Forwarding to Foreign Network**
The encapsulated packet is sent through the network to the foreign network where the mobile node is currently located.

• **Decapsulation**
The foreign agent (or the mobile node itself) receives the tunneled packet and removes the outer header to retrieve the original packet.

• **Delivery to Mobile Node**
The original packet is then delivered to the mobile node in the foreign network.

• **Reply from Mobile Node**
When the mobile node sends a reply, it usually sends it directly to the correspondent node using standard IP routing.

Q25. Briefly explain architecture of WLAN with diagram.

A Wireless Local Area Network (WLAN) allows devices to connect and communicate wirelessly within a limited area such as a home, office, or campus. The architecture of WLAN defines how different components interact to provide network connectivity.

The WLAN architecture mainly consists of the following components:

• **Stations (STAs)**
These are wireless devices such as laptops, smartphones, or tablets that connect to the network.

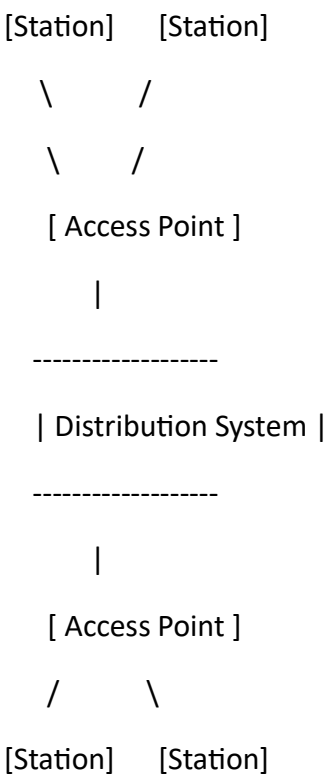
• **Access Point (AP)**
An access point is a central device that connects wireless devices to the wired network. It acts as a bridge between wireless clients and the network infrastructure.

• **Basic Service Set (BSS)**
It is a group of stations that communicate with each other through a single access point.

• **Distribution System (DS)**
This is the backbone network that connects multiple access points. It is usually a wired network.

• **Extended Service Set (ESS)**
It is formed by connecting multiple Basic Service Sets through a distribution system, allowing wider coverage.

Diagram (Conceptual Representation)



Q26. Explain 5G mobile network features in detail.

5G is the fifth generation of mobile communication technology, designed to provide faster speeds, lower latency, and support for a large number of connected devices. It is an advanced version of previous generations like 4G and is built to handle modern communication needs such as IoT, smart cities, and real-time applications.

The key features of 5G are explained below:

- **High Data Speed**

5G offers extremely high data transfer speeds, reaching up to several gigabits per second. This allows faster downloading, streaming, and data transmission.

- **Low Latency**

Latency in 5G is very low, often less than 1 millisecond. This enables real-time communication, which is essential for applications like online gaming, remote surgery, and autonomous vehicles.

- **Massive Device Connectivity**

5G can support a very large number of devices simultaneously. This is important for the Internet of Things (IoT), where many devices need to be connected at the same time.

• **Improved Reliability**
5G networks provide more stable and reliable connections, reducing the chances of call drops and network interruptions.

• **Higher Bandwidth**
It provides greater bandwidth, allowing more data to be transmitted at the same time without congestion.

• **Energy Efficiency**
5G technology is designed to be more energy-efficient, reducing power consumption in devices and network infrastructure.

• **Network Slicing**
5G supports network slicing, which allows a single physical network to be divided into multiple virtual networks, each optimized for different applications.

Q27. Explain how DHCP works in a mobile network context.

Dynamic Host Configuration Protocol (DHCP) is used to automatically assign IP addresses and network configuration parameters to devices in a network. In a mobile network context, DHCP plays an important role in assigning temporary IP addresses to mobile devices as they move between different networks.

The working of DHCP follows a standard process known as the DORA process:

• **Discover**
When a mobile device connects to a network, it does not have an IP address. It sends a broadcast message called DHCP Discover to find available DHCP servers.

• **Offer**
The DHCP server responds with a DHCP Offer message, which contains an available IP address along with other configuration details such as subnet mask, gateway, and DNS server.

• **Request**
The mobile device replies with a DHCP Request message, indicating that it accepts the offered IP address.

• **Acknowledgment**
The DHCP server sends a DHCP Acknowledgment message confirming the assignment. The device can now use the assigned IP address for communication.

In a mobile environment, when a device moves from one network to another, it must request a new IP address using the same process. The assigned IP address is usually temporary and valid for a specific lease time.

DHCP simplifies network management by automatically configuring devices and ensures efficient use of IP addresses, especially in dynamic mobile environments.

Q28. Explain Bluetooth technology with a neat figure of its protocol stack.

Bluetooth is a short-range wireless communication technology used to exchange data between devices such as smartphones, laptops, headphones, and other personal devices. It operates in the 2.4 GHz ISM band and is mainly used for Wireless Personal Area Networks (WPAN).

Bluetooth enables devices to communicate without physical cables and supports low power consumption, making it suitable for portable devices.

The Bluetooth protocol stack is organized into different layers, each performing a specific function.

- **Radio Layer**

This is the lowest layer responsible for transmitting and receiving data using radio signals.

- **Baseband Layer**

It manages physical channels, handles data packets, error correction, and timing control.

- **Link Manager Protocol (LMP)**

This layer is responsible for link setup, authentication, encryption, and power control between Bluetooth devices.

- **Logical Link Control and Adaptation Protocol (L2CAP)**

It provides multiplexing, segmentation, and reassembly of data packets. It allows multiple higher-level protocols to share the same Bluetooth link.

- **RFCOMM**

It provides serial port emulation and supports data transfer similar to traditional serial communication.

- **Application Layer**

This is the top layer where user applications such as file transfer, audio streaming, and device control operate.

Q29. List out types of congestion control and explain any one.

Congestion control is used to regulate the flow of data in a network to avoid congestion, which occurs when too much data is sent and the network becomes overloaded.

The types of congestion control are as follows:

- Open Loop Congestion Control
- Closed Loop Congestion Control

Now, one type is explained in detail.

Closed Loop Congestion Control

Closed loop congestion control is a technique where the network continuously monitors its condition and takes corrective actions when congestion is detected. It uses feedback from the network to control the flow of data.

In this method, the sender adjusts its transmission rate based on information received from the network. If congestion occurs, the network informs the sender, and the sender reduces the data transmission rate. Once the congestion is reduced, the sender gradually increases the transmission rate again.

This approach includes techniques such as congestion detection, feedback transmission, and rate adjustment. It helps in maintaining efficient network performance and prevents packet loss.

Q30. Explain Address Resolution Protocol briefly.

Address Resolution Protocol (ARP) is used to find the physical (MAC) address of a device when its IP address is known. It works within a local network.

When a device wants to communicate, it sends a broadcast ARP request asking for the MAC address of the destination IP. The device with that IP address replies with its MAC address. The sender then uses this MAC address to send data.

ARP helps in mapping IP addresses to MAC addresses for proper communication within a network.

Q36. What is HTTPS? List important key features of HTTPS.

HTTPS (HyperText Transfer Protocol Secure) is a secure version of HTTP used for communication over the Internet. It ensures that the data exchanged between a user's browser and a web server is encrypted and protected from unauthorized access.

The key features of HTTPS are as follows:

- **Data Encryption**
HTTPS uses SSL/TLS to encrypt data, preventing attackers from reading sensitive information.

- **Data Integrity**
It ensures that the data is not altered or modified during transmission.

- **Authentication**
HTTPS verifies the identity of the website using digital certificates, ensuring that users connect to a legitimate server.

- **Secure Communication**
It provides a secure channel for transmitting sensitive data such as passwords, credit card details, and personal information.

- **Protection Against Attacks**
It protects against common attacks like man-in-the-middle attacks and eavesdropping.

- **Trust and Reliability**
Websites using HTTPS are considered more trustworthy, often indicated by a lock symbol in the browser.

- **SEO Advantage**
Search engines prefer HTTPS-enabled websites, improving their ranking.

Q39. List out types of applications uses mobile computing.

Mobile computing is used in various fields where communication and data access are required on the move. The main types of applications are as follows:

- **Business Applications**
Used for mobile banking, online transactions, stock trading, and enterprise resource planning.

- **Healthcare Applications**
Used for remote patient monitoring, telemedicine, and accessing medical records.

• **Educational Applications**
Used for e-learning, online classes, digital libraries, and virtual classrooms.

• **Entertainment Applications**
Used for mobile gaming, video streaming, music, and social media platforms.

• **Location-Based Services**
Used for GPS navigation, maps, ride-sharing, and location tracking.

• **E-Commerce Applications**
Used for online shopping, mobile payments, and digital wallets.

• **Emergency and Public Safety Applications**
Used by police, fire services, and disaster management teams for communication and coordination.

Q40. Explain use of Gateways and list types of Gateways.

A gateway is a network device that connects different networks and enables communication between them, especially when they use different protocols or architectures. It acts as an entry and exit point for data between networks.

The use of gateways is explained as follows:

• **Protocol Conversion**
Gateways convert data from one protocol to another so that different networks can communicate.

• **Interconnection of Networks**
They connect heterogeneous networks such as LAN, WAN, and mobile networks.

• **Data Routing and Forwarding**
Gateways determine the path and forward data to the correct destination.

• **Security Control**
They provide security by filtering traffic, controlling access, and protecting networks from unauthorized access.

• **Data Translation**
Gateways convert data formats and ensure compatibility between systems.

Types of Gateways

• **Network Gateway**
Connects different networks and manages traffic between them.

• **Application Gateway**
Operates at the application layer and provides services like proxy and firewall.

• **Cloud Gateway**
Connects local systems to cloud services.

• **IoT Gateway**
Connects IoT devices to the Internet and manages communication between them.

• **VoIP Gateway**
Converts voice data between traditional telephone networks and IP networks.

Q41. Draw and explain architecture of mobile computing.

The architecture of mobile computing defines how mobile devices communicate with servers through wireless networks. It consists of three main components: mobile devices, communication networks, and fixed infrastructure.

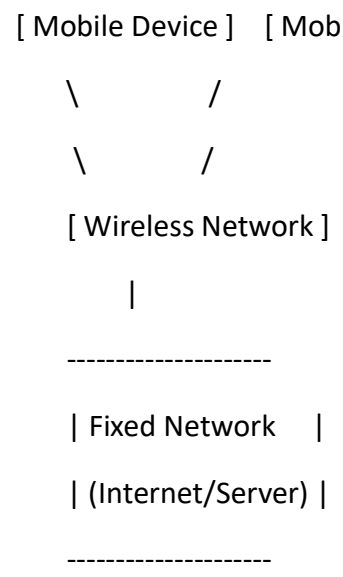
The architecture is explained below:

• **Mobile Devices**
These are portable devices such as smartphones, tablets, and laptops. They act as clients and are used by users to access services and send requests.

• **Communication Network**
This includes wireless networks such as cellular networks, Wi-Fi, or satellite systems. It enables communication between mobile devices and the fixed network.

• **Fixed Infrastructure**
This consists of servers, databases, and wired networks that process user requests and provide services such as web access, email, and data storage.

Diagram (Mobile Computing Architecture)



Q42. List security standards in mobile computing.

Security standards in mobile computing are used to protect data, ensure privacy, and provide secure communication over wireless networks.

The main security standards are as follows:

- **WEP (Wired Equivalent Privacy)**
An early security standard used in wireless networks. It provides basic encryption but is now considered weak.

- **WPA (Wi-Fi Protected Access)**
An improved version of WEP that provides better encryption and security mechanisms.

- **WPA2**
A stronger security standard that uses advanced encryption (AES) and is widely used in modern wireless networks.

- **WPA3**
The latest Wi-Fi security standard that provides enhanced protection against attacks and improved encryption.

- **SSL (Secure Sockets Layer)**
A protocol used to secure communication over the Internet by encrypting data.

• **TLS (Transport Layer Security)**
An improved version of SSL that provides stronger security for data transmission.

• **IPsec (Internet Protocol Security)**
A protocol used to secure IP communication by authenticating and encrypting data packets.

Q43. Explain key functions of communication gateway.

The key functions of a communication gateway are as follows:

• **Protocol Conversion**
The gateway converts data from one communication protocol to another, allowing different systems to interact.

• **Data Format Translation**
It translates data into a format that the receiving system can understand.

• **Message Routing**
The gateway forwards data to the correct destination based on network addresses.

• **Security Management**
It provides security features such as authentication, encryption, and filtering of data.

• **Connection Management**
The gateway establishes, maintains, and terminates communication sessions between systems.

• **Data Filtering**
It filters unwanted or unauthorized data to protect the network.

• **Interoperability Support**
It enables communication between heterogeneous systems such as mobile devices, servers, and legacy systems.

Q44. Explain use of middleware and list types of middleware.

Middleware is a software layer that acts as an interface between applications and the underlying network or operating system. It simplifies communication and data management in distributed and mobile computing environments.

The use of middleware is explained as follows:

• **Simplifies Communication**
Middleware hides the complexity of network communication and provides a simple interface for applications.

• **Enables Interoperability**
It allows different systems, platforms, and programming languages to communicate with each other.

• **Manages Data Exchange**
Middleware handles the transmission, formatting, and delivery of data between systems.

• **Provides Security**
It offers services like authentication, authorization, and encryption for secure communication.

• **Supports Scalability**
Middleware allows systems to grow and add new components without affecting existing ones.

• **Ensures Reliability**
It provides features like error handling and message buffering to maintain reliable communication.

Types of Middleware

• **Remote Procedure Call (RPC)**
Allows a program to execute functions on a remote system as if they were local.

• **Message-Oriented Middleware (MOM)**
Enables communication through message passing between systems.

• **Object Request Broker (ORB)**
Allows objects to communicate across networks in distributed systems.

• **Database Middleware**
Provides access and interaction between applications and databases.

• **Transaction Processing Middleware**
Manages transactions to ensure data consistency and integrity.

Q51. Answer the following questions:

1) List Uses of Ad hoc Network

- Used in disaster management where no fixed infrastructure is available
- Used in military communication for secure and quick deployment
- Used in temporary networks such as meetings and conferences
- Used in remote areas where network setup is difficult
- Used for emergency rescue operations
- Used in sensor networks for data collection
- Used in vehicular communication systems

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2) Enlist entities and terminology of mobile computing

- **Mobile Node (MN)** – A device that can move from one network to another
- **Home Network** – The original network of the mobile node
- **Foreign Network** – The network where the mobile node is currently located
- **Home Agent (HA)** – A router in the home network that tracks the mobile node
- **Foreign Agent (FA)** – A router in the foreign network that provides services to the mobile node
- **Care-of Address (CoA)** – A temporary address representing the mobile node's current location
- **Correspondent Node (CN)** – The device communicating with the mobile node

Q. Give difference between Client-Server and Peer-to-Peer network.

Basis	Client-Server Network	Peer-to-Peer Network
Structure	Centralized system with a dedicated server	Decentralized system with no central server
Control	Server controls resources and access	Each node controls its own resources
Role of Devices	Clients request services, server provides services	All devices act as both client and server
Security	More secure due to centralized control	Less secure due to lack of central control
Performance	Better performance under heavy load	Performance decreases as number of nodes increases
Cost	Higher cost due to server and maintenance	Lower cost, no dedicated server required
Management	Easy to manage and maintain centrally	Difficult to manage as network grows

Q60. Define ARP and RARP protocols.

• **ARP (Address Resolution Protocol)**
ARP is a protocol used to map an IP address to its corresponding physical (MAC) address within a local network. When a device knows the IP address of another device but not its MAC address, it sends a broadcast request. The device with the matching IP address replies with its MAC address.

• **RARP (Reverse Address Resolution Protocol)**
RARP is a protocol used to obtain the IP address of a device when its MAC address is known. It was mainly used by diskless machines to get their IP address from a server during startup. RARP requires a dedicated server and is now obsolete, replaced by DHCP.

Q64. Compare Classful and Classless notations.

Basis	Classful Addressing	Classless Addressing (CIDR)
Definition	IP addresses are divided into fixed classes (A, B, C)	IP addresses are assigned without fixed classes
Subnet Mask	Default subnet mask is fixed for each class	Subnet mask is flexible and can be customized
Address Allocation	Inefficient, often leads to wastage of IP addresses	Efficient allocation with minimal wastage
Routing	Does not support route aggregation effectively	Supports route aggregation (CIDR)
Flexibility	Less flexible due to fixed boundaries	Highly flexible for network design
Usage	Used in early networking systems	Used in modern networks
Example Format	Based on class (e.g., 192.x.x.x → Class C)	Uses prefix notation (e.g., /24, /16)

Q71. Explain Ad-hoc network in detail.

An Ad-hoc network is a wireless network without any fixed infrastructure, where devices communicate directly with each other. Each node acts as both a sender and a router.

Data is transmitted using multi-hop communication, where intermediate nodes forward data if the destination is not in direct range. The network is self-configuring and can change dynamically as nodes move.

Key Characteristics

• **Infrastructure-less**
No central server or access point is required.

• **Dynamic Topology**
Nodes can join or leave anytime, changing network structure.

• **Multi-hop Communication**
Data is forwarded through intermediate nodes.

• **Distributed Control**
All nodes share responsibility for network operation.

• **Self-Configuring**
Network forms automatically without manual setup.

Ad-hoc networks are used in disaster recovery, military communication, and temporary networks.

However, they face issues like security risks, limited bandwidth, and higher power consumption.

Q79. Write short note on Location Management.

Location management is the process of tracking and maintaining the current location of a mobile user in a mobile network. It ensures that calls and data can be delivered to the correct device as it moves from one area to another.

It mainly consists of two operations:

• **Location Update**
When a mobile device moves from one location area to another, it informs the network about its new location. This update is stored in network databases.

• **Paging**
When there is an incoming call or data for the mobile device, the network searches for the device in its last known location area by sending paging messages.

Location management helps in efficient use of network resources and ensures continuous communication without interruption.